Muhammad **Ayyan**

AI ENGINEER · MACHINE LEARNING

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Summary

AI Engineer with experience in designing and deploying AI agents, RAG systems, and intelligent applications using advanced LLM frameworks. Skilled in deep learning (Transformers, CNNs, NLP), data preprocessing, and building scalable production-ready solutions with Python, FastAPI, and Docker. Proficient in MLOps practices, including Apache Airflow, MLflow, CI/CD, and AWS, with a passion for developing multimodal AI systems that solve real-world problems.

Tools & Technologies

- **Languages & Libraries:**
  Python – SQL - Langchain - LangGraph - Tensorflow - Scikit-learn – NumPy – Pandas – Matplotlib – FastAPI - Pydantic - MLflow - Optuna - Hugging Face
- **AI & Machine Learning & MLOps:**
  Agents- RAG - LLMs - Fine-Tuning - Transformer - NLP - ANNs-CNNs-RNNs-LSTM - Supervised & Unsupervised Learning - Hyperparameter Tuning – Model Evaluation – EDA – Feature Engineering – Apache Airflow – Docker – CI/CD - EC2, S3
- **Tools & Platforms:**
  VSCode – Jupyter Notebook – Git – GitHub - Dagshub – MySQL - Ollama

projects

Agentic RAG System | Github | DEMO

- Developed **"Indus Guardian**," an advanced **Agentic RAG** system utilizing a **FastAPI** backend and **graph-based** routing to orchestrate intelligent multi-document retrieval and reasoning.
- Built document pipelines for parsing and indexing data in **Pinecone** for **semantic** search, plus a responsive full-stack UI for smooth user experience.
- Designed a responsive **full-stack frontend featuring** an optimized layout **architecture** with refined component spacing to deliver a seamless user interaction experience.
  Tools: **Python**, FastAPI, **LangChain**, **LangGraph**, Pinecone, HTML/CSS, Gemini API , MYSQL, GitHub, Docker,**HuggingFace-Spaces** ,JSON .

Vecihal Insurance Prediction | GitHub | DEMO

- Architected an end-to-end MLOps pipeline for vehicle insurance data, implementing automated data ingestion from **MongoDB** Atlas alongside robust custom logging and **exception-handling** modules. .
- Developed modular data validation and **transformation** components governed by **YAML schemas,** executing pipeline-ready feature engineering to streamline model training and evaluation
- Built a **production-grade** prediction pipeline integrated with a **web API** and **UI templates** to serve real-time model **inference** endpoints.
  Tools: Python, **Transformers**, PyTorch, MLflow ,FastAPI,Skit-Learn,**MongoDB**,Hugging Face, Pandas, **FastAPI**, Docker, Git, html, css,Dagshub.

Kideny Disease Clasification | GitHub | DEMO

- Built an end-to-end Deep Learning pipeline for kidney disease classification using CNNs, achieving high-accuracy medical image predictions.
- Implemented MLOps best practices with MLflow, DVC, and a modular Flask backend to enable experiment tracking, data versioning.
- Deployed a production-ready application using Docker, and GitHub Actions CI/CD for automated integration and scalable deployment.
  Tools: **Python**, **Scikit-learn**, Pandas, Docker,**TensoFlow**,Keras,Ml-Flow,DVC,**PythonHugging Face Spaces**.

Certifications

Udemy: ML/DL With Python

View Certificate

Udemy: Agentic AI

View Certificate

Academics

Virtual University of Pakistan *Lahore, Pakistan*

B.S. in Computer Science